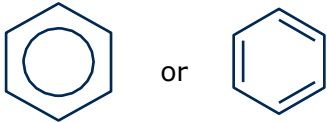


Topic 18: Organic Chemistry – Arenes

Knowledge of the common uses of organic compounds mentioned in this topic is expected.

Students will be assessed on their ability to:

18.1	be able to use thermochemical, X-ray diffraction and infrared data as evidence for the structure and stability of the benzene ring <i>Students may represent the structure of benzene as</i>  <i>as appropriate in equations and mechanisms.</i>
18.2	understand that the delocalised model for the structure of benzene involves overlap of p -orbitals to form π -bonds
18.3	understand why benzene is resistant to bromination, compared to alkenes, in terms of delocalisation of π -bonds in benzene compared to the localised electron density of the π -bond in alkenes
18.4	know the following reactions of benzene, limited to: - i oxygen in air (combustion to form a smoky flame) - ii bromine, in the presence of a catalyst - iii a mixture of concentrated nitric and sulfuric acids - iv fuming sulfuric acid - v halogenoalkanes and acyl chlorides with aluminium chloride as catalyst (Friedel-Crafts reaction)
18.5	understand the mechanism of the electrophilic substitution reactions of benzene in halogenation, nitration and Friedel-Crafts reactions, including the generation of the electrophile
18.6	understand the reaction of phenol with bromine water and the reasons for the relative ease of this reaction compared to benzene
	Further suggested practical: Carry out the reactions in 18.4, and 18.6 where appropriate (using methylbenzene or methoxybenzene)